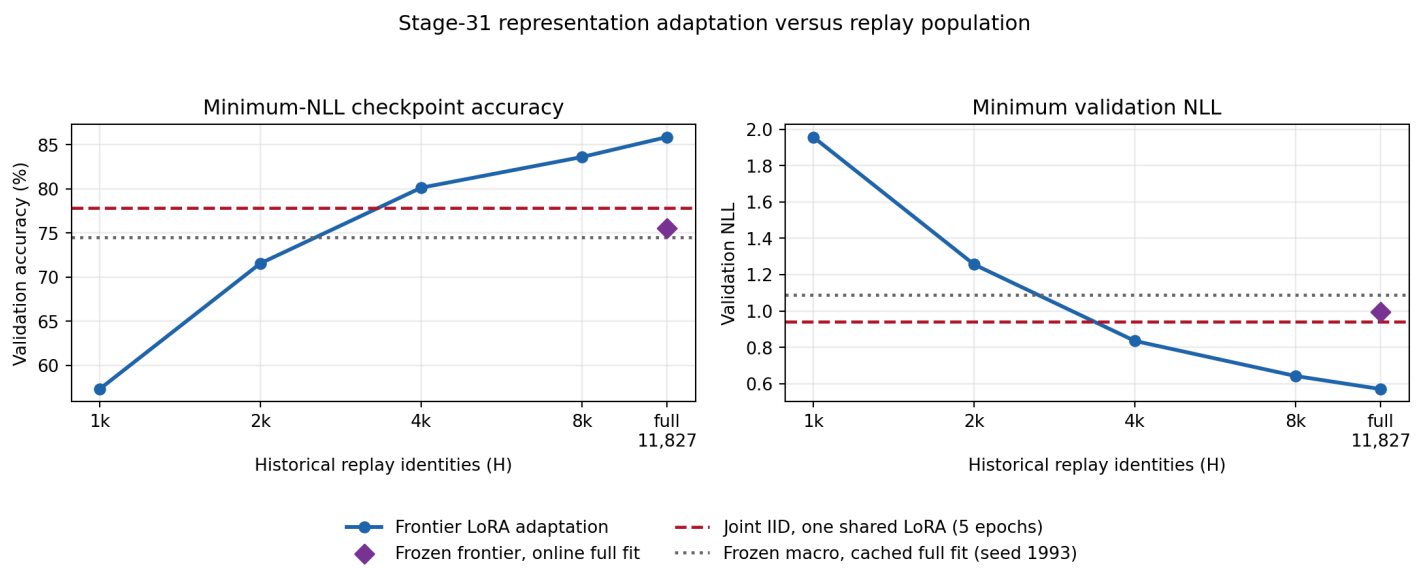


# ImageNet-R stage-31 frontier-LoRA adaptation

Can jointly adapting five fragmented node representations close the same-split joint-IID accuracy and NLL gap?

**Primary result.** The minimum-NLL adaptive condition is Frontier LoRA adaptation (H=11,827; full fit): **85.864% accuracy / 0.5710 NLL**. That is 8.003 percentage points higher and 0.3715 NLL lower than the five-epoch joint-IID reference. Frontier LoRA adaptation (H=4,096) is the smallest H to reach both joint-IID references, first doing so at epoch 6.



All blue points jointly update the five frontier LoRAs and macro head. Checkpoints minimize validation NLL. Horizontal references use the identical fit/validation identities but different architectures.

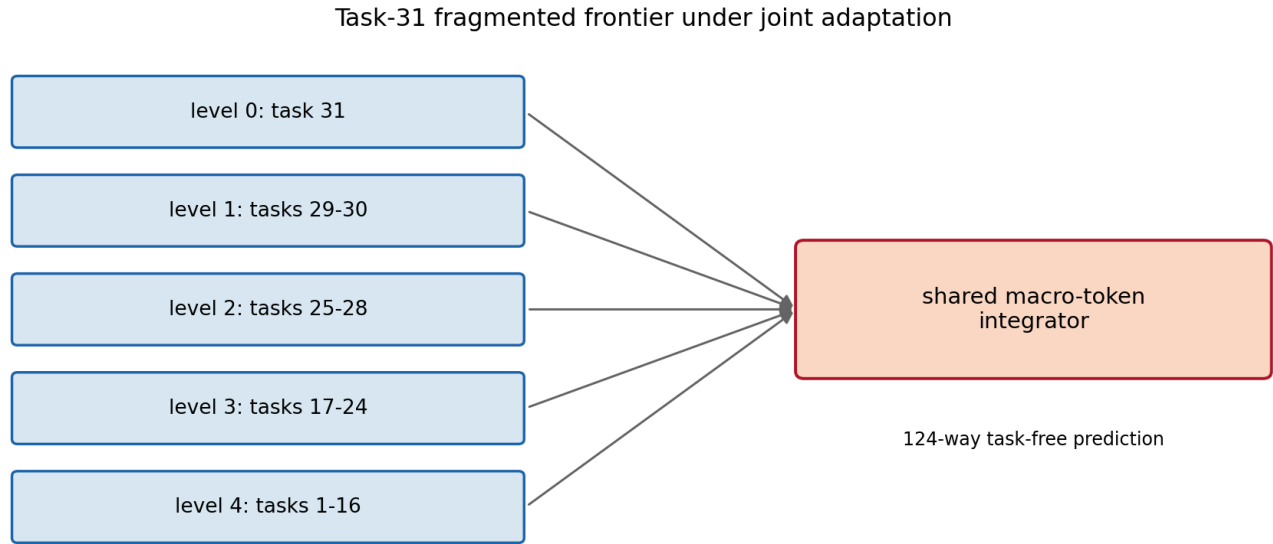
## Complete condition summary

| Condition                                     | Min-NLL epoch | Accuracy at min NLL | Minimum NLL | Maximum accuracy | NLL at max accuracy | Runtime   |
|-----------------------------------------------|---------------|---------------------|-------------|------------------|---------------------|-----------|
| Frontier LoRA adaptation (H=1,024)            | 31            | 57.330%             | 1.9572      | 57.527%          | 1.9624              | 21.9 min  |
| Frontier LoRA adaptation (H=2,048)            | 25            | 71.564%             | 1.2565      | 71.564%          | 1.2565              | 30.1 min  |
| Frontier LoRA adaptation (H=4,096)            | 12            | 80.125%             | 0.8360      | 80.846%          | 0.8443              | 47.1 min  |
| Frontier LoRA adaptation (H=8,192)            | 8             | 83.601%             | 0.6430      | 84.093%          | 0.6534              | 88.2 min  |
| Frontier LoRA adaptation (H=11,827; full fit) | 13            | 85.864%             | 0.5710      | 85.864%          | 0.5710              | 120.0 min |
| Frozen frontier, online full fit              | 16            | 75.533%             | 0.9967      | 76.418%          | 1.0063              | 43.4 min  |

Maximum accuracy and its NLL are reported separately to expose calibration tradeoffs. Every cell also includes all 367 current-task identities; maximum historical H=11,827 is exactly the 12,194-image full fit.

## Architecture and experimental boundary

The task-31 frontier has five nodes at levels 0-4, covering task intervals 31, 29-30, 25-28, 17-24, and 1-16. Each node supplies its own final 197 x 768 LoRA-adapted token sequence and immutable local affine scores. The macro transformer combines them without task IDs or labels.



Each arrow carries 197 adapted 768-value tokens plus local affine scores.

The base ViT and node classifiers are frozen. Only five rank-16 LoRAs and the shared 12.06M-parameter macro head can move.

Every cell includes all 367 current-task images.  $H=1,024, 2,048, 4,096, 8,192,$  and  $11,827$  are nested prefixes of one deterministic uniform draw from the historical tasks without replacement. Maximum  $H$  therefore gives exactly all 12,194 fit identities. Every cell starts independently from identical sealed node tensors and macro initialization. The validation partition has 3,049 clean identities and never contributes gradients. No test image is opened.

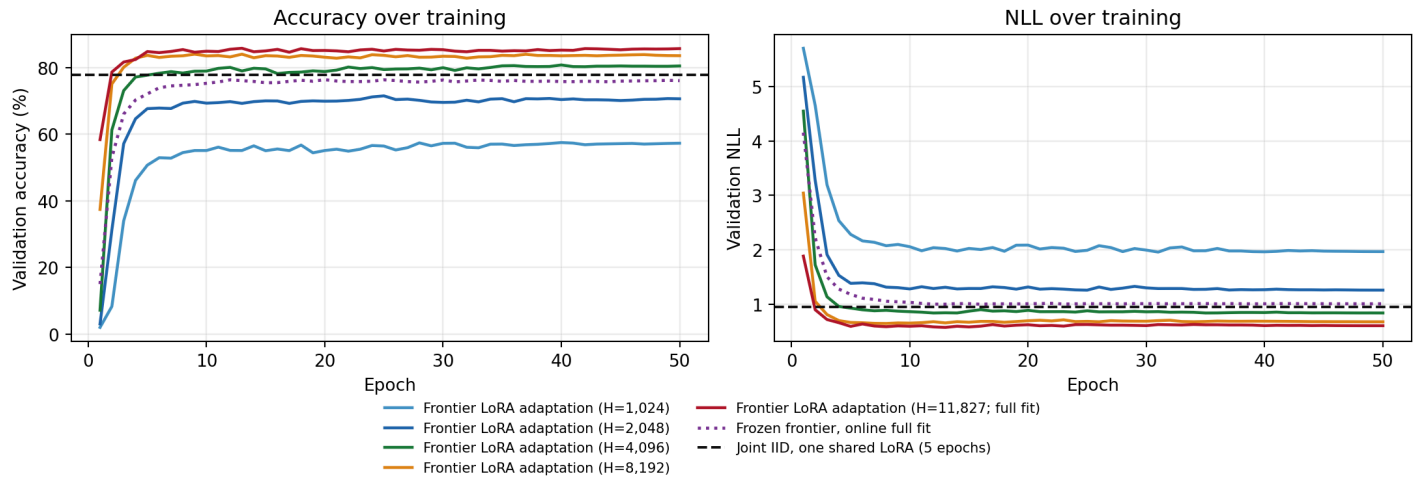
**Compute boundary.** The adaptive frontier evaluates five node-specific ViTs and the macro transformer for every image. Joint IID evaluates one ViT with one shared LoRA and classifier. Their split and full-fit data exposure match, but parameter count, training compute, and deployment compute do not.

## Why the frozen online control exists

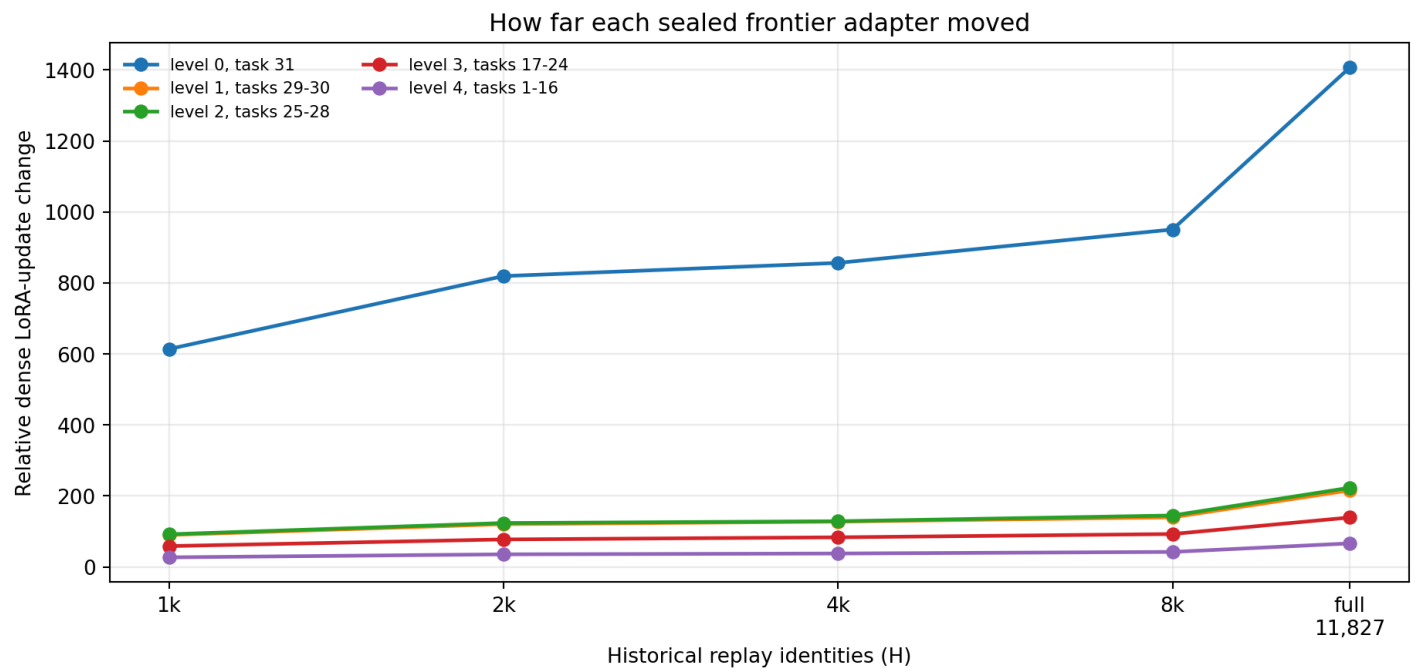
The previous frozen macro was trained from cached center-crop tokens. This run loads images online with deterministic random training augmentation. The purple full-fit control follows that new path while freezing the LoRAs, so its difference from the cached gray reference measures the pipeline/augmentation change rather than representation adaptation.

# Optimization behavior

All checkpoints; dashed black is the five-epoch joint-IID endpoint



Every epoch is retained in a hash-chained history. The dashed black line is the five-epoch joint-IID endpoint, not a stopping gate.



Scale-aware Frobenius movement of each dense LoRA update, measured from its sealed source node at the selected minimum-NLL checkpoint.

## Interpretation

Allowing the frontier LoRAs to move changes the result by 10.331 percentage points and 0.4257 NLL relative to the online frozen-LoRA control at their minimum-NLL checkpoints.  $H=4,096$  is the smallest tested historical population to cross both joint-IID values. At exactly 5 full-fit passes (60,970 image presentations), adaptive full history is 84.880% / 0.5897, frozen full history is 72.155% / 1.1788, and joint IID is 77.862% / 0.9425. This isolates feature adaptation from training exposure, but not the frontier's extra model and deployment compute.

The head schedule is the previous minimum-NLL winner: effective batch 64, peak AdamW LR  $3e-5$ , 50 epochs, 5% warmup, and cosine decay. The LoRA peak LR  $5e-4$  is imported from the same-split joint recipe but run here in AdamW under the shared schedule. It has not been tuned for this coupled model.

**Limits.** This is a one-seed validation screen. The H cells change both unique data and total optimizer updates. Maximum accuracy is exploratory because all 50 validation checkpoints are visible. A promising condition needs replication and a focused LoRA/head learning-rate audit before any locked-test use.

## Integrity and reuse

The training seal records zero fit-validation overlap, zero test overlap, zero test evaluations, and unchanged source hierarchy files. A second pass authenticated all six completed cells without constructing their five ViTs or taking an optimizer step. Large checkpoints and model weights remain local; compact histories, tables, plots, and protocol records are the report surface.